STAT340 - Topic 2: Monte Carlo methods

Monte Carlo

- Monte Carlo (MC) is a class of **simulation-based** methods for computation, with wide applications in math/science
- Stanisław Ulam (who worked at both UW-Madison & the Manhattan Project) first applied MC to computing the probability a random game of solitaire is solvable
- Origin of name: supposedly named after his uncle's favorite casino
- *Very* flexible approximate method, can do many things, e.g:
  - find probabilities
  - compute expectations
  - estimate parameters
  - test hypotheses,
  - etc…



MC probability estimates

Let E be some event for an RV X. Suppose we want to estimate $\Pr[E]$

Define the **indicator function** $\mathbf{1}_E$ where

$$\mathbf{1}_E(x) = \begin{cases} 1 & \text{if } x \in E \\ 0 & \text{if } x \notin E \end{cases}$$

Let $X_1, \ldots, X_n$ be iid from same distribution as X, and let $Y_i = \mathbf{1}_E(X_i)$, what RV is Y_i?

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Ans: $Y_i \sim \text{Bernoulli}(p = \Pr[E])$, so that $\mathrm{E}[Y_i] = p = \Pr[E]$,

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Also by LLN, as sample size grows, we'll have $\bar{Y} \to \mathrm{E}[Y_i] = \Pr[E]$

In other words, to approximately find probability of event E, we just need to:

1. Run experiment X a large number of repetitions,
2. Count how many times E occurs,
3. Divide by number of repetitions.

Example 1: normal draws

As an easy first example, let $X \sim \text{Normal}(0, 1^2)$, and let $E = (-2, 2)$, use MC to evaluate $\Pr[E]$

```
# set MC repetitions size
M = 1e6

# draw n samples
X = rnorm(M, mean=0, sd=1)

# count how many in E
E = sum(-2<X & X<2)

# divide by n to get P(E)
E/M
```

```
[1] 0.954342
```

```
# compare with theoretical value
pnorm(2, mean=0, sd=1) - pnorm(-2, mean=0, sd=1)
```

```
[1] 0.9544997
```

Example 2: birthday problem

Suppose birthdays are evenly distributed throughout year (and ignore leap years). In a group of 23 people with independent birthdays, what's the probability **at least 2 people have the same birthday**?

```
# sample 23 days from 1-365 with replacement
(bdays = sample(365, size=23, replace=TRUE))
```

```
 [1] 324 138 208 114 335   8 109 168 317 303  56 142 120 192 109 330  94 108 268 319 103 316 188
```

```
# check if more than 1 person has same birthday
max(table(bdays)) > 1
```

```
[1] TRUE
```

```
# loop above using for()
M = 1e5 ; count = 0
for(i in 1:M){
  if(max(table(sample(365,23,TRUE)))>1){
    count = count+1
  }
}
count / M
```

```
[1] 0.50253
```

```
# alternative way using replicate()
mean(replicate(M, max(table(sample(365,23,TRUE)))>1 ))
```

```
[1] 0.50902
```

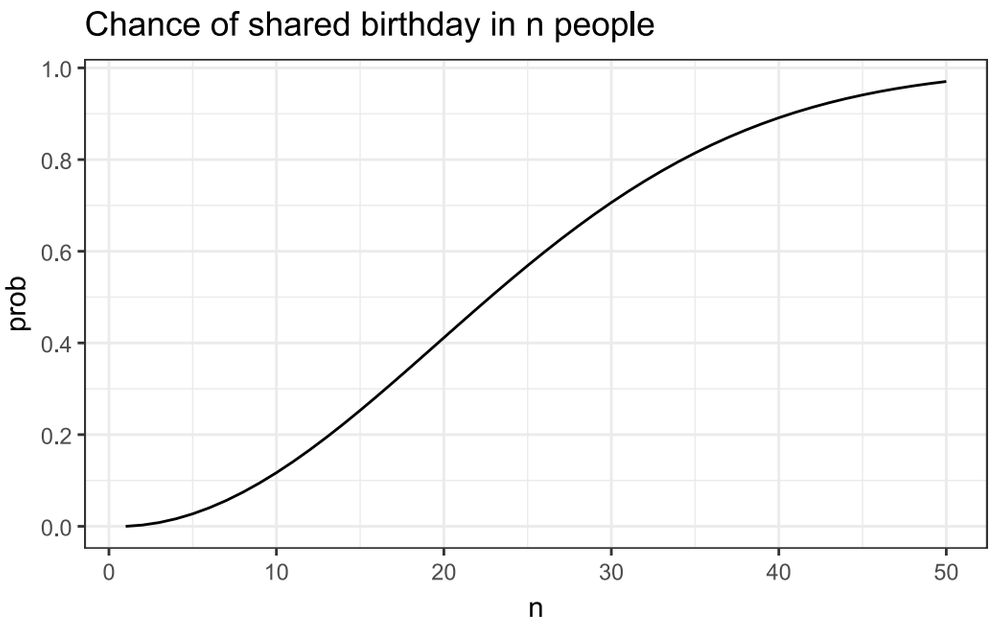
Example 2: birthday problem (cont.)

This problem is exactly solvable. For n people, $1 \leq n \leq 365$, probability all n people have distinct birthdays is:

$$1 \cdot \frac{364}{365} \cdot \frac{363}{365} \cdots \frac{365-n+1}{365} = \frac{365!}{(365-n)!\,365^n}$$

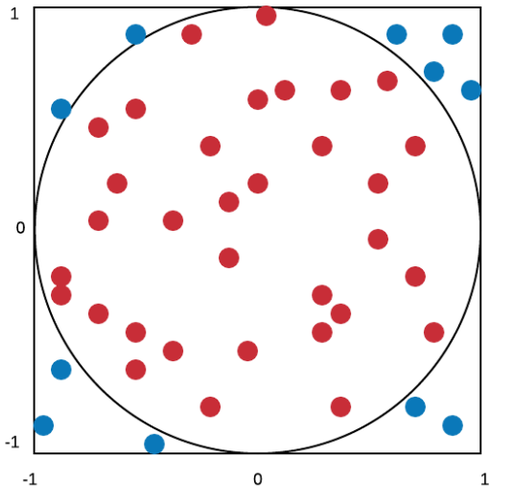
Subtracting this from 1 gives the probability at least 2 people share a birthday. We can define and plot this function in R:

```
# vectorized bday function
bday = Vectorize(
  \(n) 1-prod(365:(365-n+1)/365)
)
tibble(n=1:50, prob=bday(n)) %>%
  ggplot(aes(n,prob)) + geom_line() +
  scale_y_continuous(breaks=0:5/5) +
ggtitle("Chance of shared birthday in n people")
```



Assigned reading: 9.3.6 – Buffon's needle for more practice & 9.4 – More uses for application to finding expectations in general & integration

MC parameter estimates

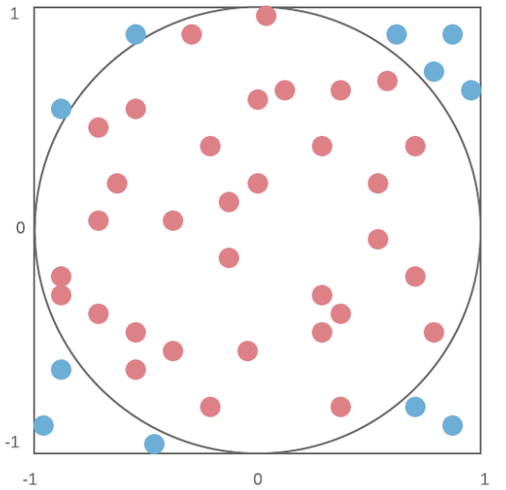


If the simulation result is related to some unknown parameter, this can be used to derive an **estimator** of the parameter.

Example: Consider the length-2 square S with corners at $(\pm 1, \pm 1)$ and its inscribed circle C with center $(0, 0)$ and radius 1.

Note a point uniformly sampled over S has $\pi/4$ probability of landing in the circle.

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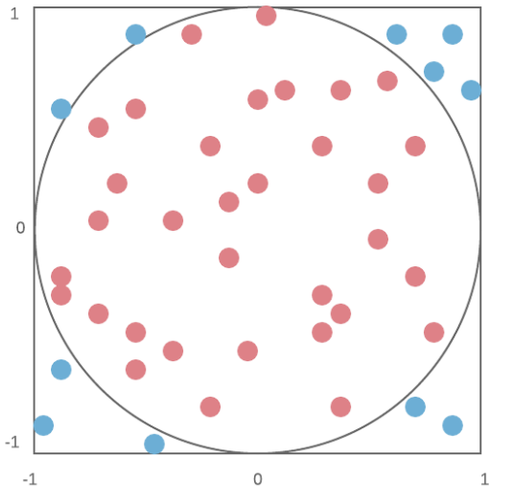
Note a point uniformly sampled over S has $\pi/4$ probability of landing in the circle.

Let (X_i, Y_i) be a point uniformly sampled over S, and let $Z_i = \mathbf{1}_C(X_i, Y_i)$ where

$$\mathbf{1}_C(x, y) = \begin{cases} 1 & \text{if } x^2 + y^2 \leq 1 \\ 0 & \text{if } x^2 + y^2 > 1 \end{cases}$$

be the indicator function for if the point is in C. Note $Z_i \sim \text{Bernoulli}(\pi/4)$

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Thus, we have the following estimator $\hat{\pi} = 4\bar{Z}$, and we know for iid sample of size n:

$$\mathrm{E}[\hat{\pi}] = 4\,\mathrm{E}[\bar{Z}] = \pi \qquad \mathrm{Var}[\hat{\pi}] = 4^2\,\mathrm{Var}[\bar{Z}] = \frac{16}{n}\left(\frac{\pi}{4}\right)\left(1 - \frac{\pi}{4}\right) = \frac{\pi(4-\pi)}{n}$$

$\hat{\pi}$ example in R

```
# set sample size
n = 1e6

# simulate X, Y points uniformly over square
X = runif(n, min=-1, max=1)
Y = runif(n, min=-1, max=1)

# how many points also inside circle?
C = sum(X^2 + Y^2 <= 1)

# produce estimate
(est = 4 * C / n)
```

```
[1] 3.140784
```

```
# percent error
(est-pi)/pi*100
```

```
[1] -0.02574024
```

Validating estimator performance

We saw previously $\mathrm{E}[\hat{\pi}] = \pi$ and $\mathrm{Var}[\hat{\pi}] = \pi(4-\pi)/n$, do these hold?

Pick $n = 1000$, i.e. 1000 points will be used to produce each estimate, and repeat the process to get $M = 100$ different MC estimates, then check their mean/var:

```
# use replicate() to get estimates
(est = replicate(100,4*mean(runif(1e3,-1)^2+runif(1e3,-1)^2<1)))
```

```
  [1] 3.136 3.128 3.084 3.092 3.148 3.104 3.184 3.188 3.168 3.068 3.068 3.224 3.132 3.136 3.192
 [16] 3.112 3.044 3.124 3.092 3.108 3.156 3.172 3.060 3.060 3.200 3.188 3.124 3.080 3.092 3.100
 [31] 3.136 3.144 3.108 3.064 3.220 3.200 3.260 3.104 3.168 3.176 3.068 3.096 3.080 3.084 3.028
 [46] 3.132 3.144 3.116 3.140 3.124 3.040 3.188 3.180 3.212 3.192 3.132 3.176 3.236 3.132 3.104
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 [76] 3.308 3.152 3.120 3.200 3.136 3.300 3.172 3.084 3.136 3.124 3.132 3.064 3.120 3.148 3.088
 [91] 3.096 3.180 3.172 3.128 3.148 3.136 3.244 3.156 3.104 3.116
```

```
# get mean/var of estimates
c(mean(est), var(est))
```

```
[1] 3.142160000 0.003198196
```

```
# mean clearly agrees with theory, how about var?
pi*(4-pi)/1000
```

```
[1] 0.002696766
```

Simulating RVs with arbitrary PDFs

So far, we've only learned how to use built-in `r...()` functions or `sample()` to simulate variables, but can we simulate from arbitrary PDF? Yes, using the following fact:

For any continuous RV X with CDF F_X, we have

$$F_X(X) \sim \text{Uniform}(0, 1)$$

Note it's $F_X(X)$ not $F_X(x)$, i.e. the CDF is applied to an observation drawn from X.

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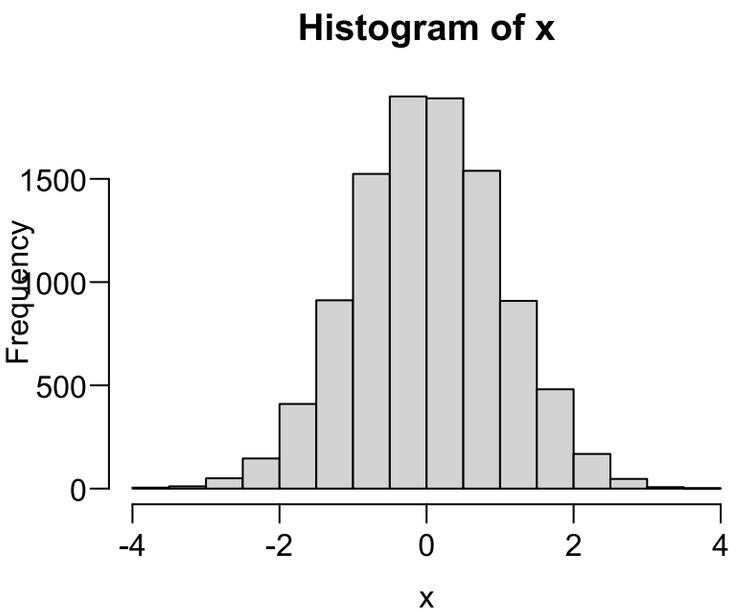
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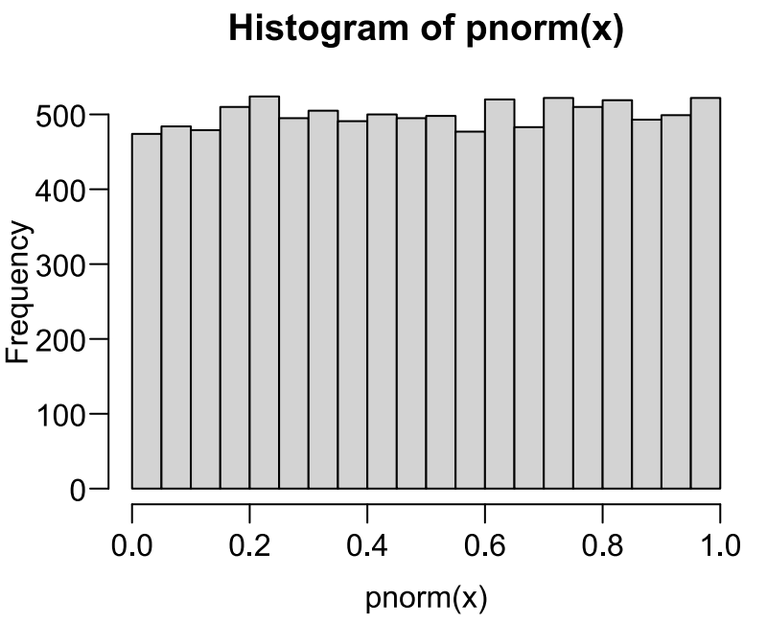
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Put another way: percentiles of continuous RV draws are uniform. Demo:

```
# start with norm
x = rnorm(1e4)
hist(x)
hist(pnorm(x))
```





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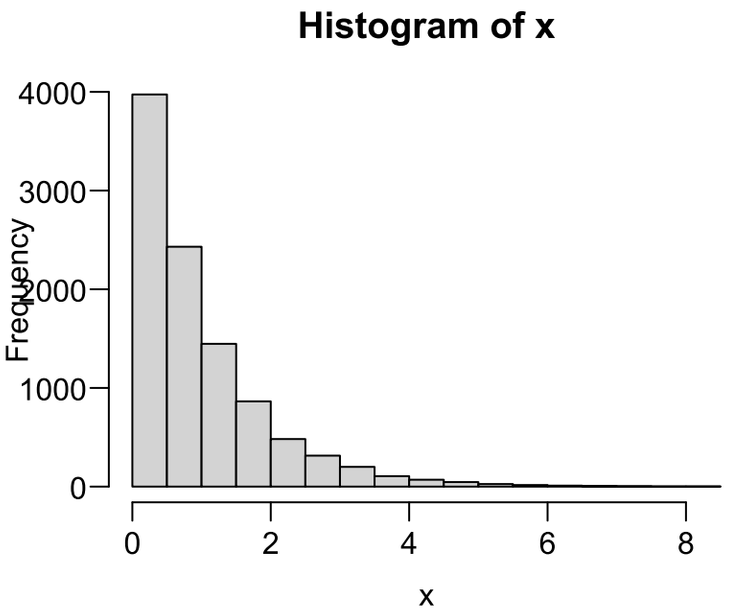
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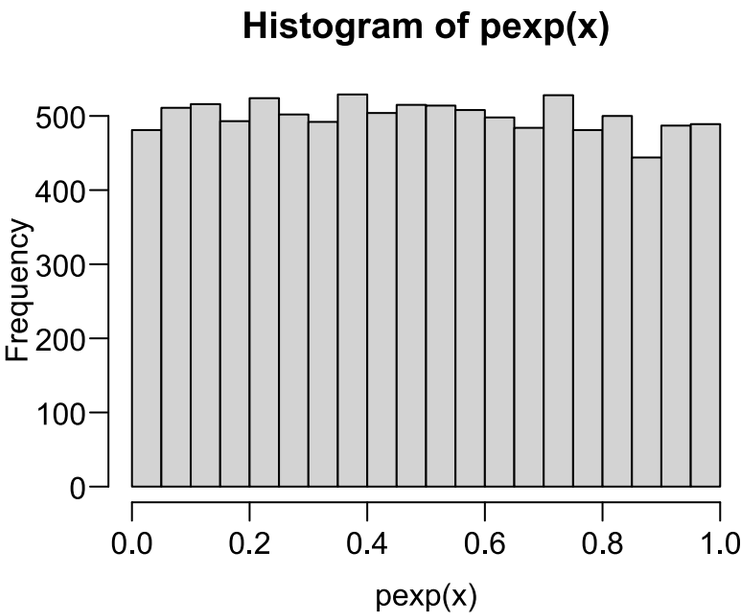
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```
# now try exp
x = rexp(1e4)
hist(x)
hist(pexp(x))
```





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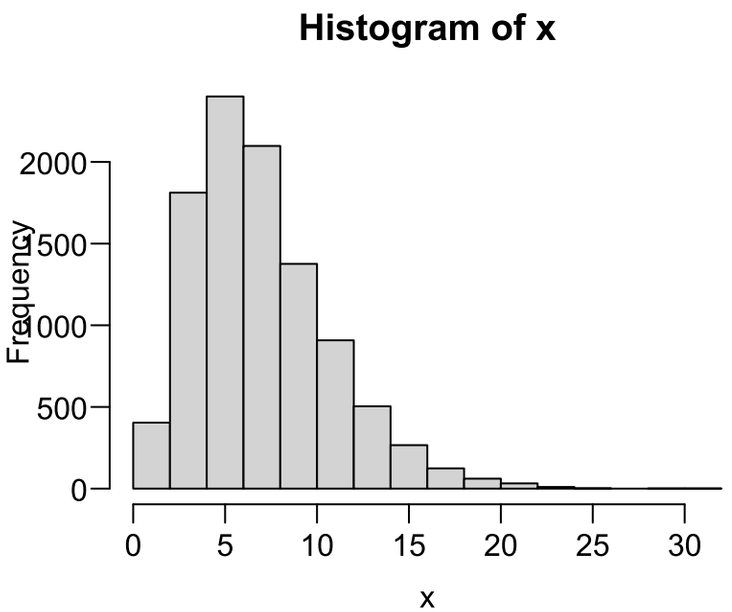
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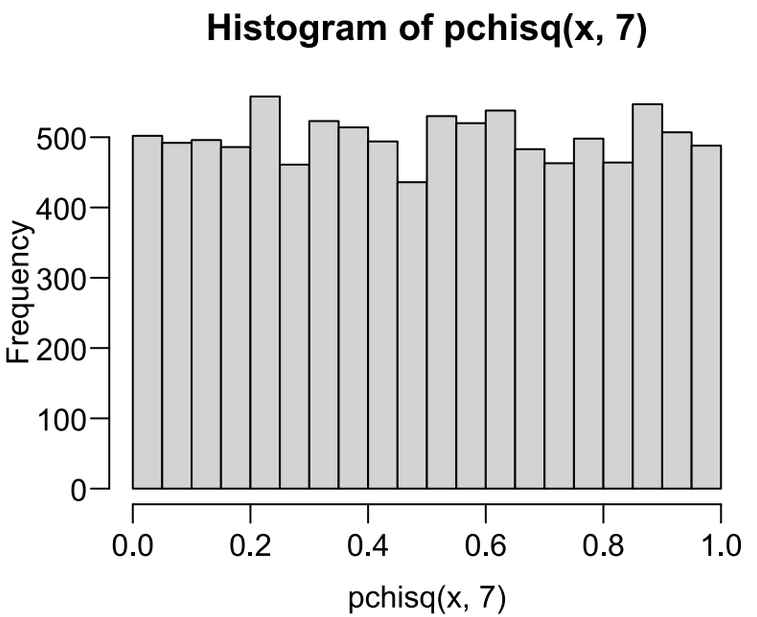
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Put another way: percentiles of continuous RV draws are uniform. Demo:

```r
# some other dists
x = rchisq(1e4,7)
hist(x)
hist(pchisq(x,7))
```





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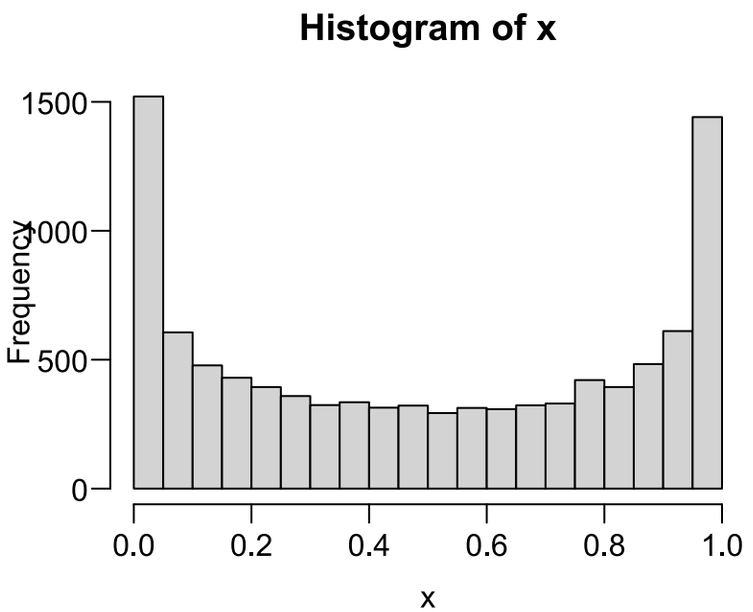
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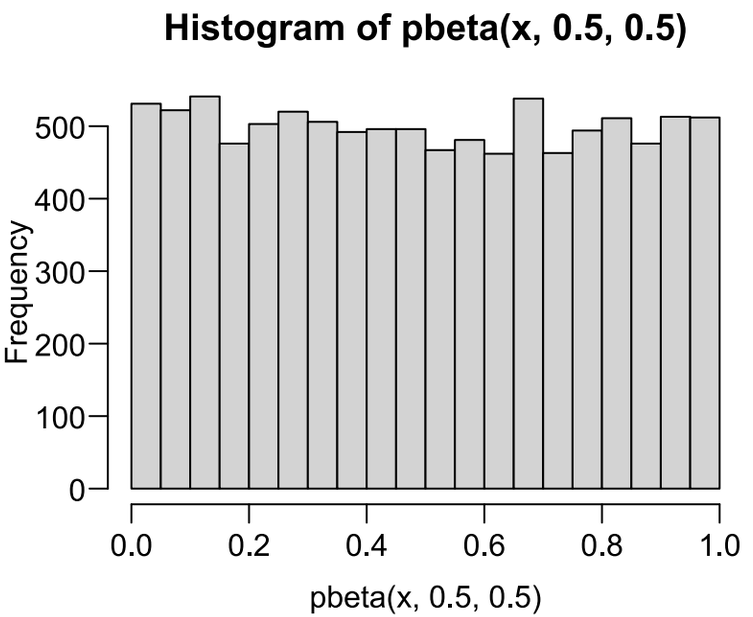
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Put another way: percentiles of continuous RV draws are uniform. Demo:

```
# some other dists
x = rbeta(1e4,.5,.5)
hist(x)
hist(pbeta(x,.5,.5))
```





$F_X(X)$ uniformity

Quick proof of this fact. Let $Y = F_X(X)$, i.e. Y is the percentile of a draw from X. Consider the CDF of Y, $F_Y(y)$. By definition:

$$F_Y(y) = \Pr[Y \leq y] = \Pr[F_X(X) \leq y]$$

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By monotonicity of CDFs, $F_X(X) \leq y$ implies $F_X^{-1}(F_X(X)) \leq F_X^{-1}(y)$, so

$$\Pr[F_X(X) \leq y] = \Pr[X \leq F_X^{-1}(y)] = F_X(F_X^{-1}(y)) = y$$

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$$\Pr[F_X(X) \leq y] = \Pr[X \leq F_X^{-1}(y)] = F_X(F_X^{-1}(y)) = y$$

This shows $F_Y(y) = y$ which is uniquely the CDF of $\mathrm{Uniform}(0, 1)$.

Brian's notes section 9.7 has a nice visual showing this for each draw

Generating from arbitrary PDFs

Since $F_X(X) \sim \text{Uniform}(0, 1)$ for any continuous X, it follows that

$$X \sim F_X^{-1}(\text{Uniform}(0, 1))$$

I.e. to simulate an X with arbitrary PDF, we can run its quantile function on $\text{Uniform}(0.1)$

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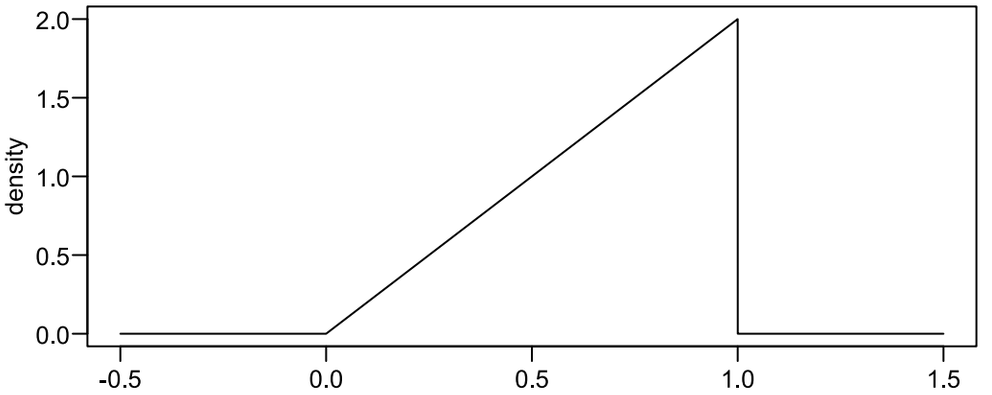
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Exercise: simulate 1000 draws from PDF:

$$f(x) = \begin{cases} 2x & 0 \leq x \leq 1 \\ 0 & \text{otherwise} \end{cases}$$

with CDF $F(x) = \int_0^x f(x)dx = x^2$




```
# plot pdf
density = \(x) ifelse(0<x & x<1, 2*x, 0)
plot(density, from=-.5, to=1.5)
```

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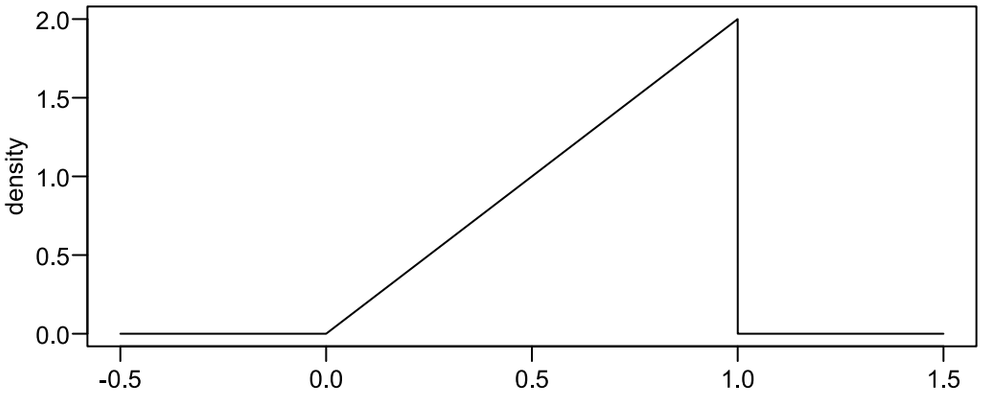
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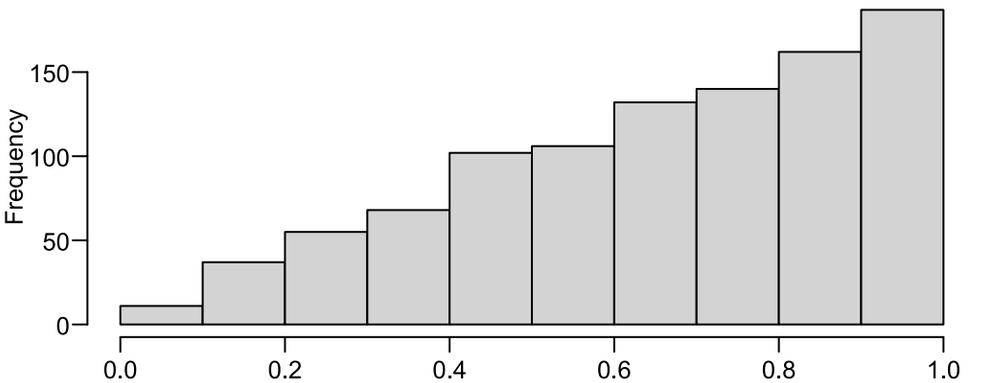
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```



Ans: apply inverse, i.e. `sqrt()`, to `runif()`

```
x = sqrt(runif(1000))
hist(x)
```



Note on randomness

- For MC, it's important to have a good source of randomness, which is surprisingly hard
- In R, random number generation is not truly random but **pseudorandom**
- Starting with a seed value, a scrambling algorithm called Mersenne Twister is used to imitate randomness (if no seed set, system time + process ID used)

```
# set seed and generate
set.seed(1)
rnorm(8)
```

```
[1] -0.6264538  0.1836433 -0.8356286  1.5952808  0.3295078 -0.8204684  0.4874291  0.7383247
```

```
# reset seed and regenerate same values
set.seed(1)
rnorm(8)
```

```
[1] -0.6264538  0.1836433 -0.8356286  1.5952808  0.3295078 -0.8204684  0.4874291  0.7383247
```

- Useful for reproducibility in MC problems
- For true randomness, usually need to measure something, e.g. atmospheric noise, radioactive decay, or even lava lamps (used by Cloudflare)